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Integrated Master's student specializing in Visual AI and Deep Learning, with a First-Class academic track record (74% avg). Proven experience in open-source engineering and implementing advanced architectures for computer vision and temporal modelling.

EDUCATION

- **University of Bristol** 2022-date
Computer Science MEng Average Percentage (Years 1-3): 74
 - **Notable modules:** Applied Deep Learning (73), Advanced Visual AI (69), Image Processing and Computer Vision (71), Machine Learning (77), Data-Driven Computer Science (81), Mathematics for Computer Science B (90), Object Oriented Programming and Algorithms 1 (76), Algorithms 2 (75), Advanced Algorithms (80), Team Project (80)
 - **Masters Dissertation:** The feasibility of using AI models to assist first time buyers in the housing market (ongoing)
 - * **Supervisor:** James Cussens
 - * **Scope:** Investigating how the temporal data of a housing area may be used in the development and comparison of Forest and Transformer based methods to help first time buyers gain purchase confidence
 - Competed in the 2023 BrisHack 24-hour Hackathon in a team of 5
 - Competed in the unitaryHACK 2025 open source quantum Hackathon in a team of 2
- **Norwich School** 2020-2022
A levels Results: A*A*A*A
 - Computer Science (A*), Maths (A*), Further Maths (A*), Physics (A)
- **Norwich School** 2017-2020
GCSEs Results: 9 subjects ranging 7-9

TECHNICAL SKILLS

- **Languages:** Python, Java, C#, Swift, Go, C
- **Deep Learning and AI:** PyTorch, Hugging Face Transformers, Tensorflow, OpenCV, Scikit-learn, Pandas, NumPy
- **Engineering & Tools:** Git, AWS, CI/CD, Unit Testing

ACADEMIC PROJECTS

- **Comparative Analysis of Deep Learning Methodologies for Single-Image Super-Resolution (75)** (Code and Report)
 - Evaluated GANs (BSRGAN), Diffusion models (SR3), and Transformers (SwinIR) on the DIV2K dataset
 - Demonstrated that Transformer models outperform GANs and Diffusion in noise robustness and LPIPS metrics (0.55 vs 0.64/0.56)
 - Analysed computational efficiency, identifying Diffusion models as computationally expensive (6,020 GFLOPs) compared to Transformers (19.77 GFLOPs)
 - Investigated the impact of various scaling and noise degradation techniques on model performance
- **Siamese ProgressionNet for Temporal Ordering in Egocentric Video (79)** (Code and Report)
 - Developed a Siamese network to determine pairwise temporal ordering in cooking videos using the HD-EPIC dataset
 - Conducted studies on data augmentation and learning rates, improving F1 score by 19% through cropping and perspective techniques
 - Implemented layer freezing strategies to analyse regularization effects and prevent overfitting on small dataset samples
- **Comparative Analysis of Statistical Learning & Stochastic Models (76)** (Code and Report)
 - Introduced PCA to reduce feature space (54 to 41 dimensions) while retaining 95% variance, and evaluated the efficacy of K-Means and Gaussian Mixture Models (GMM) on high-dimensional, non-independent ecological data
 - Demonstrated the limitations of SVMs on non-linearly separable data and contrasted this with the robustness of Random Forests, which achieved 95.5% accuracy compared to 67% for Logistic Regression
 - Conducted Bayesian Linear Regression by defining priors and sampling from posterior distributions, comparing results against a custom Multi-Layer Perceptron (MLP) to demonstrate the susceptibility of NNs to overfitting on small datasets (MSE 0.23 vs 0.06)

OPEN SOURCE

- **Internalised NomicBERT model for the HuggingFace/Transformers repository** (PR: #43067)
 - Integrated RoPE directly into the self-attention mechanism
 - Replaced standard GELU with SwiGLU to enhance model stability and performance
 - Implemented gradient checkpointing and key-value caching to optimise training memory efficiency and accelerate autoregressive generation
 - Developed weight conversion script to map original Nomic tensors to the Hugging Face architecture
 - Wrote a comprehensive test suite covering unit tests, integration tests, and generation consistency (handling left-padding and attention masks), resolving complex edge cases in batch inference
 - Wrote model documentation describing the model and use cases
- **Introduced qubit routing documentation for the open source quantum circuit optimisation repository PYZX** (PR: #304)

APPLIED ENGINEERING PROJECTS

- **Interactive Museum Exhibit (77)** (Code)
 - Engineered a visually interactive exhibit for Bristol Museum as part of a team of 5, utilising .NET and Avalonia frameworks, alongside implementing unit tests and CI/CD practices
- **End-to-End ANPR Gate System** (Code)
 - Created an ANPR gate system for an industrial park, utilising OpenCV, Matplotlib and Pytesseract to extract number plates and determine access levels for park entry
 - Developed a PHP/MySQL web interface for real-time user management and database logging
- **Real-Time Sepsis Prediction Model**
 - Developed a support tool to predict sepsis from live medical features.
 - Trained a transformer model with a recall of 0.75 and AUC of 0.74, and a LightGBM model with a recall 0.82 and AUC 0.9
- **Computer Vision Stop-Sign Detector** (Code)
 - Implemented a traffic sign detector from scratch without high-level CV libraries, utilising custom implementations of Sobel edge detection and Hough Space transforms to understand fundamental vision algorithms.
- **Distributed Game of Life Simulation (77)**
 - Engineered a high-performance distributed simulation capable of concurrent processing across local cores or AWS instances, implementing parallel execution strategies relevant to large-scale computing.

INTERESTS AND VOLUNTEERING

- **Charity:** Planned and directed a 24-hour walk for Help for Heroes, raising £500 *2018*
- **C2 World's Longest Tandem Row:** 7 days straight and over 1,100,000m *2020*
- **Volunteering:** During A-levels, coached on weekends to teach kids how to row *2021-2022*
- **Rowed at Henley Royal Regatta:** Represented both school and university *2021-2023*

REFERENCES

- **Dr Steven Ramsay**
 - Associate Professor in Programming Languages, University of Bristol
 - **Email:** steven.ramsay@bristol.ac.uk
- **Dr James Cussens**
 - Senior Lecturer, University of Bristol
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